Work sheet-I for the Course Econometrics

1. Discuss the main difference between economic model and econometric model
2. Outline the four steps involved in econometrics and indicate the most important contents of each step.
3. What is the meaning of stochastic relationship and how do you specify it?
4. Econometrics deals with the measurement of economic relationships which are stochastic or random. The simplest form of economic relationships between two variables X and Y can be represented by:

   $Y_i = \beta_0 + \beta_1 X_i + U_i$; Where β_0 *and* β_1 = are regression parameters and U_i = the stochastic disturbance term. What are the reasons for the insertion of U-term in the model?
5. Researcher is using data for a sample of 10 observations to estimate the relation between consumption expenditure and income. Preliminary analysis of the sample data produces the following data.

   $\sum xy = 700$, $\sum x^2 = 1000$, $\sum X = 100$ $\sum Y = 200$ $\sum y^2 = 1400$

   Where $x = X_i - \overline{X}_i$ and $y = Y_i - \overline{Y}$

   a. Use the above information to compute OLS estimates of the intercept and slope coefficients and interpret the result
   b. Compute the value of **R^2** (coefficient of determination) and interpret the result
   c. Compute 95% confidence interval for the slope parameter
   d. Test the significance of the slope parameter at 5% level of confidence using t-test
6. Assume the following hypothetical weekly data on Y (Demand for a normal good) and X (its price) are obtained from a certain market

Y_t	1	3	5	6	5	6	4	7	8	7	9
X_t	7	6	5	4	4	4	4	3	3	4	2

The relationship between Y and X is expressed as;

$$Y_t = \beta_0 + \beta_1 X_t + e_t$$

Where, t is time representing weeks in which the data are collected

A. State the assumptions of the classical linear regression model to obtain each desirable properties of OLS slope of two-variable model.
B. Assuming the model satisfies all the assumptions of the classical linear regression model estimate the parameters, their standard errors, and goodness of fit.
C. Test the statistical significance of the parameter estimates, $\hat{\beta}_0$ and $\hat{\beta}_1$ (H_0: $\hat{\beta}_i$ = 0) at 1 percent level of significance
D. Construct a 95% confidence interval for the true slope of the demand function

7. The following data refers to the price of a good 'P' and the quantity of the good supplied, 'S'.

P	2	7	5	1	4	8	2	8
S	15	41	32	9	28	43	17	40

   a. Estimate the linear regression line $\mathrm{E}(S) = \alpha + \beta P$
   b. Estimate the standard errors of $\hat{\alpha}$ *and* $\hat{\beta}$
   c. Test the hypothesis that price influences supply
   a. Obtain a 95% confidence interval for α
   b. If in a 9th market the rate of price is p = 2, predict the quantity supply (S) in this market.